

Introduction

This project focuses on analysing a sales dataset to enhance inventory management practices for businesses. As a data analyst, the goal is to extract valuable insights from the data, including sales trends, stock levels, and product performance. The analysis will help identify top-selling products, low-stock items, and seasonal sales patterns, ultimately providing actionable recommendations to optimize inventory management. The project will involve data preprocessing, exploratory data analysis (EDA), and the creation of visualizations to communicate key findings. The final deliverable will include a report with insights and recommendations to help businesses improve their inventory efficiency.

Description of the Dataset Analysis Approach and Methodology

The analysis of the sales dataset will follow a structured approach to ensure a comprehensive understanding of sales trends, stock levels, and product performance, ultimately leading to actionable inventory management recommendations.

1. Data Collection and Preprocessing

- **Data Acquisition:** The sales dataset will be obtained from the provided source, containing sales transactions, product details, stock levels, and timestamps.
- **Data Cleaning:** Preprocessing steps will include handling missing values, correcting inconsistencies, and ensuring proper data types for each attribute (e.g., numerical values, dates).

2. Exploratory Data Analysis (EDA)

- **Sales Trends:** Sales performance will be analysed over time, looking at daily, weekly, or monthly sales patterns. This will help identify trends, seasonality, and fluctuations in demand.
- **Top-Selling Products and Categories:** A detailed analysis will be performed to identify the most popular products, categories, and their contribution to overall sales. This will include examining the number of units sold, revenue generated, and product performance in different regions.
- **Stock Levels and Low-Stock Items:** The stock levels will be analysed to identify products that are low in stock, potentially leading to stockouts. These insights will be vital for making replenishment decisions.

3. Inventory Insights and Recommendations

- **Key Performance Indicators (KPIs):** Key metrics such as inventory turnover, stock-to-sales ratio, and reorder points will be calculated to assess inventory performance.

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- **Stock Efficiency:** The analysis will determine if the inventory levels are optimal, identifying overstocked or understocked items, and recommend adjustments to reduce carrying costs and meet customer demand.
- **Demand Forecasting:** Basic forecasting methods (e.g., moving averages) will be applied to predict future demand for products, helping to better align stock levels with anticipated sales.

4. Data Visualization

- **Sales Trends and Insights:** Interactive visualizations will be created to display trends over time, highlighting sales patterns, product performance, and inventory metrics.
- **Inventory Metrics:** Visualizations such as bar charts and heatmaps will be used to present key metrics (e.g., stock-to-sales ratio, low-stock items) in an easily digestible format.
- **Comparative Analysis:** Charts comparing sales across products, categories, and time periods will help visualize performance differences and aid in decision-making.

5. Documentation and Reporting

- A detailed report will summarize the findings from the analysis, including insights on sales trends, popular products, stock levels, and inventory performance.
- Actionable recommendations will be provided to optimize inventory management practices, such as adjusting reorder points, reducing overstock, or increasing stock for high-demand products.

This methodology ensures a systematic and data-driven approach to inventory management, resulting in valuable insights that can help businesses optimize stock levels and improve operational efficiency.

Exploratory Data Analysis (EDA)

A. Analyze Sales Trends Over Time

- Objective: Understand how sales vary over months or years.
- Approach:
 - Use ReleaseYear (or other time-related columns if available).
 - Aggregate SoldCount by time (e.g., monthly/yearly).

Visualization: Line chart for sales trends.

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B. Identify Top-Selling Products

- Objective: Find products that generate the most sales.
- Approach:
 - Group data by SKU_number and sum SoldCount.
 - Sort products by sales volume to get the top performers.
- Visualization: Bar plot of top 10 products.

C. Stock Level Analysis

- Objective: Identify low-stock items.
- Approach:
 - Focus on ItemCount to find products with critically low stock.
 - Combine with sales data to prioritize restocking.

Visualization: Bar plot or scatter plot.

D. Analyze ItemCount and SoldCount

- **Objective:** Understand the relationship between stock levels and sales.
- **Approach:**
 - Analyze how the number of items in stock (ItemCount) correlates with the number of items sold (SoldCount).
 - Calculate **Stock Left** ($\text{ItemCount} - \text{SoldCount}$) and **Sell-through Rate** ($\text{SoldCount} / \text{ItemCount}$).
 - Identify products with high sales but low stock and vice versa.
- **Visualization:**
 - **Bar plot** for Stock Left per product.
 - **Scatter plot** to show the relationship between ItemCount and SoldCount.

Source Code for EDA and data Processing

1.Data Processing :

```
import pandas as pd

# Load your dataset

data = pd.read_csv("../Inventory-Dataset.csv")

data.head()

data.info()
# Check for missing values in the dataset

missing_values_summary = data.isnull().sum()


# Display columns with missing values

print("Missing values in the dataset:")

print(missing_values_summary[missing_values_summary > 0])
# Check and Remove Duplicates

data.drop_duplicates(inplace=True)
# Convert 'File_Type' and 'MarketingType' to lowercase for consistency

data['File_Type'] = data['File_Type'].str.lower()

data['MarketingType'] = data['MarketingType'].str.lower()
# Fill missing values in 'SoldFlag' and 'SoldCount' columns with 0

data['SoldFlag'].fillna(0, inplace=True)

data['SoldCount'].fillna(0, inplace=True)

print("Missing values in 'SoldFlag' and 'SoldCount' have been filled with 0.")
# Rechecking for missing values in the dataset

missing_values_summary = data.isnull().sum()


# Display columns with missing values

print("Missing values in the dataset:")

print(missing_values_summary[missing_values_summary > 0])
```

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#Handling values with 0 in PriceReg and LowUserPrice column

```
import pandas as pd
```

```
# Replace 0 values in 'PriceReg' with the mean
```

```
data['PriceReg'] = data['PriceReg'].replace(0, data['PriceReg'].mean())
```

```
data['LowUserPrice'] = data['LowUserPrice'].replace(0, data['LowUserPrice'].mean())
```

```
#Converting Datatype for ReleaseYear to Int for calculations
```

```
data['ReleaseYear'] = data['ReleaseYear'].astype(int)
```

```
# Save the cleaned dataset
```

```
cleaned_file_path = 'InventoryDataset-Final-Cleaned.csv'
```

```
data.to_csv(cleaned_file_path, index=False)
```

```
print(f"Cleaned dataset with non-zero 'PriceReg' saved to: {cleaned_file_path}")
```

2.Exploratory Data Analysis (EDA) Source code :

```
import pandas as pd
```

```
# Load your dataset
```

```
data = pd.read_csv("../InventoryDataset-Final-Cleaned.csv")
```

```
data.head()
```

```
#Exploratory Data Analysis (EDA)
```

```
#A. Sales Trends Over Time
```

```
import matplotlib.pyplot as plt
```

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Aggregate sales by year

```
sales_trends = data.groupby('ReleaseYear')['SoldCount'].sum().reset_index()
```

Plot sales trends

```
plt.figure(figsize=(5, 3))
```

```
plt.plot(sales_trends['ReleaseYear'], sales_trends['SoldCount'], marker='o', color='b')
```

```
plt.title('Sales Trends Over Time', fontsize=16)
```

```
plt.xlabel('Release Year', fontsize=12)
```

```
plt.ylabel('Total Sales (SoldCount)', fontsize=12)
```

```
plt.grid(True)
```

```
plt.xticks(sales_trends['ReleaseYear'], rotation=45)
```

```
plt.show()
```

#B. Top-Selling Products

Aggregate sales by SKU_number

```
top_products = data.groupby('SKU_number')['SoldCount'].sum().reset_index()
```

```
top_products = top_products.sort_values(by='SoldCount', ascending=False).head(10)
```

Plot top-selling products

```
plt.figure(figsize=(12, 6))
```

```
plt.bar(top_products['SKU_number'].astype(str), top_products['SoldCount'], color='orange')
```

```
plt.title('Top 10 Best-Selling Products', fontsize=16)
```

```
plt.xlabel('SKU Number', fontsize=12)
```

```
plt.ylabel('Total Sales (SoldCount)', fontsize=12)
```

```
plt.xticks(rotation=45)
```

```
plt.show()
```

C. Identify low-stock products (stock < 10 but not 0)

```
low_stock_items = data[(data['ItemCount'] < 10) & (data['ItemCount'] > 0)] # Filter for products with stock between 0 and 10
```

```
low_stock_items_sorted = low_stock_items.sort_values(by='ItemCount')
```

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```
# Plot low-stock items

plt.figure(figsize=(12, 6))

plt.bar(low_stock_items_sorted['SKU_number'].astype(str),
low_stock_items_sorted['ItemCount'], color='red')

plt.title('Low Stock Items', fontsize=16)

plt.xlabel('SKU Number', fontsize=12)

plt.ylabel('Stock Level (ItemCount)', fontsize=12)

plt.xticks(rotation=45)

plt.show()
```

D. Correlation between ItemCount and SoldCount

```
import pandas as pd

import matplotlib.pyplot as plt

import seaborn as sns

# EDA: Correlation between ItemCount and SoldCount

correlation = data[['ItemCount', 'SoldCount']].corr()

print("\nCorrelation between ItemCount and SoldCount:")

print(correlation)

# Visualizations:

# Scatter plot to show relationship between ItemCount and SoldCount

plt.figure(figsize=(8, 5))

sns.scatterplot(data=data, x='ItemCount', y='SoldCount')

plt.title('ItemCount vs SoldCount')

plt.xlabel('ItemCount')

plt.ylabel('SoldCount')

plt.show()
```

Inventory Insights and Recommendations

To provide **Inventory Insights and Recommendations**, we can calculate key performance indicators (KPIs) like **inventory turnover**, **stock-to-sales ratio**, and **reorder points**. These metrics will help in evaluating the current inventory management system and suggest improvements based on sales patterns.

1. Inventory Turnover

The **inventory turnover ratio** measures how many times inventory is sold and replaced over a period. It is calculated using the formula:

$$\text{Inventory Turnover} = \frac{\text{Total Sold Count}}{\text{Average Inventory}}$$
$$\text{Inventory Turnover} = \frac{\text{Total Sold Count}}{\text{Average Inventory}}$$

Where:

- **Total Sold Count** is the total number of units sold.
- **Average Inventory** is the average number of items in stock over a period.

Key Insight:

- A high inventory turnover ratio suggests efficient inventory management, where products are selling quickly.
- A low ratio could indicate slow-moving inventory, which may need to be discounted or removed.

2. Stock-to-Sales Ratio

- The **stock-to-sales ratio** measures the relationship between the stock level and the sales volume. It shows how much inventory is available for each unit sold.
- $\text{Stock-to-Sales Ratio} = \frac{\text{Item Count}}{\text{Sold Count}}$
 $\text{Stock-to-Sales Ratio} = \frac{\text{Sold Count}}{\text{Item Count}}$
- A high stock-to-sales ratio could indicate overstocking, while a low ratio may suggest potential stockouts.

Key Insight:

- If the ratio is too high, it may indicate that stock is sitting idle, tying up capital.
- A low ratio may point to understocking, increasing the risk of stockouts.

3. Reorder Points

The **reorder point** is the inventory level at which a new order should be placed to avoid stockouts. It can be calculated using the formula:

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$\text{Reorder Point} = \text{Average Daily Sales} \times \text{Lead Time}$
 $\text{Reorder Point} = \text{Average Daily Sales} \times \text{Lead Time}$

Where:

- **Average Daily Sales** is the average number of units sold per day.
- **Lead Time** is the number of days it takes to receive new stock after placing an order.

Key Insight:

- If inventory levels fall below the reorder point, it's time to reorder stock to avoid stockouts.

Source Code for Inventory Insights

1. Inventory Turnover

```
#Inventory Turnover
# Calculate total sales and average inventory
total_sales = data['SoldCount'].sum()
average_inventory = data['ItemCount'].mean()
# Calculate inventory turnover ratio
inventory_turnover = total_sales / average_inventory
print(f"Inventory Turnover Ratio: {inventory_turnover:.2f}")
```

2. Stock-to-Sales Ratio

```
#Stock-to-Sales Ratio
# Calculate stock-to-sales ratio, avoid division by zero
data['Stock_to_Sales'] = data.apply(lambda row: row['ItemCount'] / row['SoldCount'] if
row['SoldCount'] != 0 else 0, axis=1)
# Get average stock-to-sales ratio (ignore zeros)
avg_stock_to_sales = data[data['SoldCount'] > 0]['Stock_to_Sales'].mean()
print(f"Average Stock-to-Sales Ratio: {avg_stock_to_sales:.2f}")
```

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3. Reorder Points

```
# Reorder Points

# Calculate average daily sales (assuming 30 days in a month)
average_daily_sales = data['SoldCount'].sum() / 30

# Define lead time (for example, 10 days)
lead_time = 10

# Calculate reorder point
reorder_point = average_daily_sales * lead_time

print(f"Reorder Point: {reorder_point:.2f} units")
```

Source Code for Visualizations

1. Sales by Marketing Type

```
# Sales by Marketing Type

sales_by_marketing = data.groupby('MarketingType')['SoldCount'].sum().reset_index()

# Plot sales by marketing type

plt.figure(figsize=(10, 6))

plt.bar(sales_by_marketing['MarketingType'], sales_by_marketing['SoldCount'])

plt.title('Sales by Marketing Type')

plt.xlabel('Marketing Type')

plt.ylabel('Total Sales')

plt.xticks(rotation=45)

plt.show()
```

2. New Release Products Analysis

```
#New Release Products Analysis

# Sales for new and old releases

sales_by_release = data.groupby('New_Release_Flag')['SoldCount'].sum().reset_index()
```

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```
# Plot sales for new vs old releases

plt.figure(figsize=(10, 6))

plt.bar(sales_by_release['New_Release_Flag'], sales_by_release['SoldCount'])

plt.title('Sales for New vs Old Releases')

plt.xlabel('New Release Flag (0 = Old, 1 = New)')

plt.ylabel('Total Sales')

plt.show()
```

3. Inventory Turnover Visualization

```
import matplotlib.pyplot as plt

# Assuming 'data' is a pandas DataFrame with 'SoldCount' and 'ItemCount' columns

# Calculate total sales and average inventory

total_sales = data['SoldCount'].sum()

average_inventory = data['ItemCount'].mean()

# Calculate inventory turnover ratio

inventory_turnover = total_sales / average_inventory

# Plot the data

plt.figure(figsize=(8, 5))

# Plot Total Sales and Average Inventory

plt.bar(['Total Sales'], [total_sales], color='blue', alpha=0.6, label='Total Sales')

plt.bar(['Average Inventory'], [average_inventory], color='orange', alpha=0.6, label='Average Inventory')

# Plot Inventory Turnover Ratio as a horizontal line

plt.axhline(inventory_turnover, color='green', linestyle='--', label=f'Inventory Turnover: {inventory_turnover:.2f}')

# Add labels and title

plt.xlabel('Metrics')

plt.ylabel('Values')
```

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```
plt.title('Inventory Turnover Analysis')  
  
# Show legend  
plt.legend()  
  
# Show plot  
plt.show()  
  
# Print the calculated Inventory Turnover  
print(f"Inventory Turnover Ratio: {inventory_turnover:.2f}")
```

Recommendations for Optimizing Inventory Management

Based on the calculated KPIs—**Inventory Turnover Ratio (591.27)**, **Stock-to-Sales Ratio (45.89)**, and **Reorder Points (8164.67 units)**—here are tailored recommendations to optimize inventory management:

1. Optimize High Inventory Turnover

- **Observation:** The inventory turnover ratio is very high (591.27), indicating that products are being sold and replaced quickly.
- **Recommendations:**
 - **Maintain adequate stock levels** for fast-moving products to avoid potential stockouts.
 - Consider increasing safety stock levels for high-demand products to buffer against unexpected spikes in sales or delays in replenishment.
 - **Streamline restocking processes** to align with the high turnover rate, ensuring stock is replenished on time.

2. Address High Stock-to-Sales Ratio

- **Observation:** The Stock-to-Sales Ratio is 45.89, which suggests that inventory levels are significantly higher than the sales demand.
- **Recommendations:**
 - **Reduce overstocking:**
 - For products with slow sales, evaluate the necessity of current stock levels and consider reducing reorder quantities.

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- Use promotions, discounts, or bundle offers to clear excess stock and free up working capital.
- **Implement inventory segmentation:**
 - Identify and classify inventory into fast-moving, slow-moving, and obsolete (FSO) categories. Allocate storage and replenishment priorities based on this segmentation.
- **Forecast demand more accurately:**
 - Use historical sales data to fine-tune demand forecasts and align stock levels with expected sales.

3. Utilize Reorder Points Effectively

- **Observation:** The reorder point of 8164.67 units suggests when a new order should be placed based on the average daily sales and lead time.
- **Recommendations:**
 - Implement a **Just-In-Time (JIT)** approach for products with stable demand to ensure inventory is replenished only when needed.
 - Adjust reorder points seasonally to account for variations in demand, especially during peak sales periods.
 - Monitor **lead times** for suppliers closely. If there are delays, increase reorder points accordingly to ensure stock continuity.

Monitor Key Metrics Continuously

- Schedule regular reviews of:
 - **Inventory Turnover Ratio:** Ensure it remains in a balanced range, reflecting efficient stock usage.
 - **Stock-to-Sales Ratio:** Keep it closer to industry benchmarks to prevent overstocking or understocking.
 - **Reorder Points:** Adjust dynamically based on demand trends, lead times, and market conditions.

Expected Benefits of Recommendations

- **Improved Inventory Efficiency:** By reducing overstocking and aligning stock levels with demand, working capital tied up in inventory can be optimized.
- **Enhanced Customer Satisfaction:** Maintaining adequate stock of high-demand products minimizes stockouts, ensuring customers can always find what they need.

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- **Reduced Inventory Costs:** Cutting down excess stock and using data-driven restocking reduces holding and replenishment costs.
- **Increased Profitability:** By focusing on fast-moving and high-margin items, overall profitability can be boosted.